

NOMA-ISAC Empowered UAV Networks: Joint Clustering and Power Allocation Optimization

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Abstract—This letter proposes a matching scheme within non-orthogonal multiple access-integrated sensing and communication (NOMA-ISAC) based uncrewed aerial vehicle (UAV) networks. NOMA-ISAC technology is utilized to optimize the downlink transmission delay (TD) and conditional mutual information (MI) performance for UAVs. The user clustering, flight trajectory and power allocation of UAVs are optimized jointly. Recognizing the varying population density of users, an improved density-weighted agglomerative hierarchical clustering (DWAHC) algorithm and a dynamic load balancing algorithm are applied to assign each user to the appropriate UAV. Then, a multi-agent deep reinforcement learning algorithm is employed to optimize the UAVs' trajectory and transmission power through interactive learning. Experimental results demonstrate that the proposed scheme outperforms orthogonal multiple access (OMA) and other baseline algorithms in terms of reducing TD and improving conditional MI.

Index Terms—NOMA, ISAC, trajectory optimization, power allocation, multi-agent deep reinforcement learning.

I. INTRODUCTION

WITH the proliferation of 5G/B5G technologies, UAV-assisted networks have become indispensable for emergency communications where terrestrial infrastructure is unavailable. In disaster scenarios or large-scale events, UAVs can support base stations (BS) to ensure rapid information flow through high mobility, low cost, and line-of-sight (LoS) connectivity advantages [1]. The integration of ISAC enhances UAV capabilities by unifying radar and communication functions, enabling both real-time environmental awareness and data transmission on shared hardware platforms. Reference [2] proposes a multi-UAV cooperative sensing and transmission scheme leveraging virtual multiple-input and single-output (MISO) to minimize mission completion time. Reference [3] introduces a data rate triggered sensing-control method to optimize mmWave/THz backhaul communication and motion control.

As spectral efficiency demands grow in high density emergency cases, conventional OMA is unsustainable for dynamic user populations' sensing and communication coordination. The bottleneck motivates the transition to NOMA, which overcomes spectral inefficiency through power-domain multiplexing. Reference [4] demonstrates NOMA's superiority in dynamic

multiple-input-multiple-output (MIMO) environments through intelligent anti-jamming power coordination, achieving 128% higher spectral efficiency than OMA under dense user scenarios. The integration of NOMA with ISAC's hardware unification resolves two fundamental tradeoffs in UAV networks, supporting high density user connectivity through power domain multiplexing and enabling unified sensing and communication operations. Reference [5] proposes an UAV-assisted NOMA-ISAC framework that synergizes power domain multiplexing with full duplex sensing and communication coordination, dynamically allocating beamforming resources and trajectory under Cramér-Rao bounds (CRB) constraints.

However, the mobility of ground users makes system performance unstable. It is necessary to introduce dynamic clustering mechanism in NOMA-ISAC system. [6] proposes a weighted K-means UAV deployment strategy that integrates user geographic locations and activity-level weights, thereby optimizing service latency. In [7], agglomerative hierarchical clustering (AHC) enables adaptive user clustering in THz-NOMA systems without preset cluster numbers. On the other hand, traditional optimization methods are ineffective due to the changing channel conditions. Deep reinforcement learning (DRL) is adopted, which learns optimal policies through direct environmental interactions. In [8], a hybrid DRL architecture combining deep deterministic policy gradient (DDPG) for continuous power control and deep Q-networks (DQN) for discrete trajectory scheduling resolves non-convex resource coordination in UAV-assisted NOMA networks. Meanwhile, [9] employs multi-agent DQN to achieve dynamic subchannel and power allocation under massive connectivity constraints, demonstrating DRL's effectiveness in stochastic wireless environments. However, the existing NOMA-ISAC frameworks fail to effectively address the simultaneous optimization of dynamic user clustering, power allocation, and 3D trajectory planning under mobility, leaving significant room for improvement in both system performance and resource management.

In this letter, we propose a NOMA-ISAC empowered UAV network using multi-agent DuelingDQN. The key contributions are: **1)** We propose a multi-agent reinforcement learning method that jointly optimizes 3D UAV trajectories and power allocation to minimize TD and maximize conditional MI. **2)** We design a two-phase user clustering process: density-aware initial grouping via dynamic weighted AHC (DWAHC), followed by prioritized migration of the farthest users from overloaded clusters to adjacent underloaded clusters, ensuring load balancing under dynamic user mobility. **3)** Extensive benchmarking across performance enhancement, algorithmic efficiency, and trajectory optimization demonstrates the superiority of our scheme in ISAC performance, resource allocation, and other related aspects.

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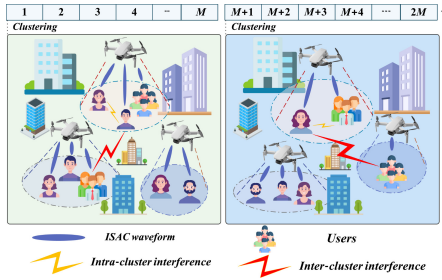


Fig. 1. System Model of NOMA-ISAC-empowered UAV Networks.

II. SYSTEM MODEL

As shown in Fig. 1, we consider a downlink scenario in a NOMA-ISAC empowered UAV network comprising K users, $k \in \mathcal{K} = \{1, \dots, K\}$ and U UAVs, $u \in \mathcal{U} = \{1, \dots, U\}$, operating over discrete time slots $t \in \mathcal{T} = \{1, \dots, T\}$. Each UAV is equipped with z antennas at position $L_u(t) = (x_u(t), y_u(t), h_u(t))$, while each user with a single antenna is located at $L_k(t) = (x_k(t), y_k(t), 0)$. Users are divided into clusters, each served by a single UAV under the NOMA-ISAC transmission scheme, where the symbols are designed to align with the corresponding UAV indices. Each UAV provides sensing and communication services to the users in its cluster. Users experience both inter-cluster and intra-cluster interferences when receiving signals. To address the issue of echo signal interference with ongoing transmissions, a multi-static architecture is adopted to separate the transmitting and receiving antennas [5]. Energy consumption is a critical bottleneck in UAV-enabled ISAC systems. Reference [10] investigates energy minimization through modeling the hovering phase, while ensuring communication and sensing quality, [11] further consider the constraint of limited return capacity. In this letter, we instead assume UAVs have sufficient on-board energy and focus solely on communication and sensing performance, omitting energy constraints [5], [12].

A. Communication Model

The channel model follows the 3GPP Release 15 specifications [13] under a discrete-time framework. For each time slot t , the channel gain between UAV u and user k is [12]:

$$g_k^u(t) = H_k^u(t) 10^{-PL_k^u(t)/10}, \quad (1)$$

where $H_k^u(t)$ is the small-scale fading coefficient, modeled as Rayleigh fading to capture rapid signal fluctuations caused by multipath propagation, $PL_k^u(t)$ denotes the link path loss, describing the average signal attenuation. The received signal at user k , $y_k^u(t)$ is:

$$y_k^u(t) = g_k^u(t) P_k^u(t) x_k^u(t) + \xi_k^u(t) + n_k^u(t), \quad (2)$$

$$\xi_k^u(t) = \xi_{inter}^u(t) + \xi_{intra}^u(t), \quad (3)$$

where $P_k^u(t)$ represents the allocated transmit power, subject to the constraint $\sum_{k=1}^{K_u} P_k^u(t) \leq P_{\max}$. $x_k^u(t)$ is the transmitting signal and $n_k^u(t) \sim \mathcal{CN}(0, \sigma^2)$ is the additive white Gaussian noise received by user k from UAV u . The interference $\xi_k^u(t)$ received by user k in cluster u , as shown in Eq. (3), consists of intra-cluster interference $\xi_{intra}^u(t)$ and inter-cluster interference $\xi_{inter}^u(t)$. The intra-cluster interference is caused by other users within cluster u , while the inter-cluster interference is generated by users in other

clusters. The specific mathematical expressions are as follows:

$$\xi_{intra}^u(t) = \sum_{m \in K_u, m \neq k} g_m^u(t) P_m^u(t) x_m^u(t), \quad (4)$$

$$\xi_{inter}^u(t) = \sum_{s=1, s \neq u}^U \sum_{n=1}^{K_s} g_n^s(t) P_n^s(t) x_n^s(t). \quad (5)$$

To reduce the influence of $\xi_{intra}^u(t)$, it is necessary to find an optimal decoding order to ensure successful successive interference cancellation (SIC) [14]. Set that cluster u contains K_u users, whose decoding order is determined by the magnitude of their equivalent channel gain $G_k^u(t)$, as: $G_k^u(t) = \frac{g_k^u(t)}{\sum_{s=1, s \neq u}^U \sum_{m=1}^{K_s} g_m^s(t) P_m^s(t) + \sigma^2}$. According to the ascending order of equivalent channel gains, the decoding priority of the users satisfies the relationship $G_{(1)}^u(t) \leq G_{(2)}^u(t) \leq \dots \leq G_{(K_u)}^u(t)$, where $G_{(1)}^u(t)$ represents the user with the smallest equivalent channel gain. For user k , all user signals with equivalent channel gains less than or equal to $G_k^u(t)$ are successfully removed via the SIC technique before decoding. However, user signals with equivalent channel gain greater than $G_k^u(t)$ will be considered noise.

Hence, the signal-to-interference-plus-noise ratio (SINR) for the k -th decoding user is shown as:

$$\gamma_k^u(t) = \frac{g_k^u(t) P_k^u(t)}{\sum_{\substack{m \in K_u \\ G_m^u(t) > G_k^u(t)}} g_m^u(t) P_m^u(t) + \sum_{\substack{s=1 \\ s \neq u}}^U \sum_{n=1}^{K_s} g_n^s(t) P_n^s(t) + \sigma^2}. \quad (6)$$

The data rate $R_k^u(t)$ between UAV u and user k , and TD $\tau(t)$ are shown as:

$$R_k^u(t) = B \log_2 \left(1 + \gamma_k^u(t) \right), \quad (7)$$

$$\tau(t) = \sum_{u=1}^U \sum_{k=1}^{K_u} \frac{D_k^u(t)}{R_k^u(t)}, \quad (8)$$

where B is the bandwidth and $D_k^u(t)$ its the amount of data transmitted by user k in cluster u .

B. Sensing Model

In this letter, conditional MI is employed to evaluate the sensing performance of UAVs. An increasing in conditional MI indicates a greater amount of information in the echo signal for target sensing, thereby enhancing the UAV's sensing capability [15]. For UAV u , the conditional MI between its sensing target impulse response and the echo signal at slot t can be succinctly characterized as [16]:

$$I_k^u(t) = \frac{1}{2} \Delta f \log_2 \left(1 + \frac{P_k^u(t) T_\nu^2 \Delta f}{\xi_k^u(t) + \sigma^2} \right), \quad (9)$$

where T_ν corresponds to the symbol duration, and Δf represents the subcarrier interval. Hence, the total conditional MI of the system is:

$$I(t) = \sum_{u=1}^U \sum_{k=1}^{K_u} I_k^u(t). \quad (10)$$

C. User Clustering Model

Effective user association is crucial in UAV-enabled networks. To address this, we propose a hierarchical two-phase

clustering framework: first, density-aware user grouping via DWAHC, followed by service load balancing through the redistribution of users from overloaded to underloaded clusters. Adaptability is ensured with periodic cluster updates at M -slot intervals, as shown in Fig. 1.

1) *DWAHC*: The density weighted distance measure, with the rest of the algorithm following the standard AHC procedure [7]. In the first stage, Gaussian kernel density estimation is applied to calculate the density distribution of user locations. The corresponding density function is given by $d(L_{target}(t)) = \frac{1}{K\eta\sqrt{2\pi}} \sum_{k=1}^K \exp(-\frac{\|L_{target}(t) - L_k(t)\|^2}{2\eta^2})$, where $L_{target}(t)$ represents the position vector of the target user, and $\|L_{target}(t) - L_k(t)\|$ is the Euclidean distance between two users, while η is the bandwidth parameter used to control the smoothness of the density estimation. The density-weighted distance $D(m, n)$ between users m and n is defined as:

$$D(m, n) = \frac{d(L_m(t)) + d(L_n(t))}{2} \|L_m(t) - L_n(t)\|. \quad (11)$$

The introduction of density-weighted distance assigns higher weights to points in high-density regions, making them more likely to cluster together, while points in low-density regions are assigned lower weights.

2) *Dynamic Load Balancing*: The second stage introduces a dynamic load balancing mechanism, featuring real-time cluster size monitoring and threshold-triggered responses. Specifically, it prioritizes the farthest users in overloaded clusters for migration to adjacent underloaded clusters until reaching preset thresholds, where iterative optimization ensures global service load balancing.

After user clustering, UAVs are deployed to each cluster. The optimization problem is transformed into a trajectory optimization and power allocation problem for the UAVs.

D. Optimization Problem of UAV

To jointly optimize the UAV flight trajectory $\mathbf{L}$ and transmit power $\mathbf{P}$ for each user group, improving matching efficiency to minimize average TD and maximize average conditional MI, the problem is formulated as:

$$\max_{\{\mathbf{L}, \mathbf{P}\}} \sum_{t=0}^T \alpha \frac{1}{\tau(t)\tau_{scale}} + (1 - \alpha) \frac{I(t)}{I_{scale}} \quad (12a)$$

$$\text{Subject to: } h_{\min} \leq h_u(t) \leq h_{\max}, \forall u, t \quad (12b)$$

$$x_{\min} \leq x_u(t) \leq x_{\max}, \forall u, t \quad (12c)$$

$$y_{\min} \leq y_u(t) \leq y_{\max}, \forall u, t \quad (12d)$$

$$L_u(t) \neq L_s(t), \forall u, s, t \quad (12e)$$

$$\sum_{k=1}^{K_u} P_k^u(t) \leq P_{\max}, \forall u, t \quad (12f)$$

$$G_{u,k}(t) \geq G_{u,m}(t), \forall u, k, t \quad (12g)$$

$$R_k^u(t) \geq R_{th}, \forall u, k, t \quad (12h)$$

$$I_k^u(t) \geq I_{th}, \forall u, k, t \quad (12i)$$

where τ_{scale} and I_{scale} are empirical parameters introduced as scaling factors to standardize the dual objectives, ensuring balanced optimization. Meanwhile, α and $1 - \alpha$

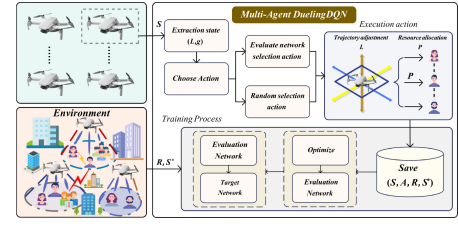


Fig. 2. Multi-Agent DuelingDQN based trajectory design and power allocation flow diagram.

denote the weighting factors for communication and sensing performance, respectively. Equations (12b)-(12d) represent the constraint on the 3D positions of UAVs, avoiding collisions or interference with other devices beyond the cellular coverage area. Equation (12e) ensures collision-free operation among UAVs. Equation (12f) sets transmission power limits to prevent UAV power consumption from exceeding the maximum transmission power. Equation (12g) establishes the decoding order for successful SIC. Equation (12h) and (12i) impose constraints on the communication rate and conditional MI to ensure compliance with service quality requirements.

Problem (12a) is a high-complexity optimization problem, primarily due to the interference from both inter-cluster and intra-cluster sources, as well as the dynamic nature of user mobility. These factors are dependent on each other, making the problem difficult to solve directly.

III. MULTI-AGENT DUELINGDQN ALGORITHMS

To solve problem (12a), we employ the multi-agent DuelingDQN algorithm to optimize UAV trajectory planning and power allocation. Each UAV acts as an autonomous agent, independently learning optimal strategies while collaborating and communicating with others. Compared to traditional DQN approaches, DuelingDQN effectively mitigates overestimation by decoupling the state value function from the action advantage function, improves the policy's generalization capability, enabling adaptive and optimal decisions in complex and dynamic environments. Fig. 2 illustrates the proposed UAV network employing the multi-agent DuelingDQN algorithm.

A. State Selection

To intuitively characterize the system state, we select the 3D coordinates of UAVs and channel gains of users as critical state representations, capturing essential spatial and link quality dynamics for reinforcement learning. The input state S is:

$$S = \{L_u(t), L_s(t), g_k^u(t), g_k^s(t)\}, \forall u, s, k, t. \quad (13)$$

B. Action Space Selection

In the multi-agent DuelingDQN framework, the action space consists of: 1) *Motion Action*: UAVs select from seven flight maneuvers: left, right, forward, backward, upward, downward, and hover. If a UAV crosses the boundary, it defaults to hovering. 2) *Power Allocation Action*: In the multi-agent DuelingDQN model, each user is assigned a set of three predefined power levels P_1, P_2, P_3 , representing the discretized power allocation options. At each decision step, the

agent selects the suitable power level for each user from their respective predefined set and maintains it until the subsequent action.

C. Network Training

In a multi-agent environment, the training network of DuelingDQN needs to adapt to collaboration and information sharing among individual agents. Each agent represents its observation information through a set of states S_u , and the action space A_u defines all possible actions of an agent. Each agent has a neural network with a DuelingDQN structure used to estimate the Q-values, it comprises a state value network and an advantage function network.

1) *Estimation of Q-Values*: For the Dueling architecture, the Q-value of taking action A_u under state S_u is composed of the state value function $V(S_u; \theta_v)$ and the advantage function $A(S_u, A_u; \theta_a)$, denoted as $Q(S_u, A_u; \theta_v, \theta_a)$.

2) *Training Process*: During the training process of each UAV, the loss function measures the difference between the squared loss of Q-values and the target Q-values. A batch of experiences (S_u, A_u, R_u, S'_u) is sampled from the experience pool based on the *batchsize*.

3) *Action Policy Selection*: Throughout the entire training process, the ϵ -greedy action policy is used to maintain a balance between exploration and exploitation.

4) *Reward Function Design*: We introduce a reward mechanism tailored for scenarios involving multi-agent collaboration. It defines the reward value as the weighted sum of the current UAV's TD and conditional MI. Additionally, to ensure satisfactory service quality for each ground user, a penalty mechanism is incorporated. The specific reward function $r(t)$ is calculated as follows:

$$r(t) = \frac{\alpha \frac{1}{\tau(t)\tau_{scale}} + (1 - \alpha) \frac{I(t)}{I_{scale}}}{2^{\max(\mathbb{I}(R_k^u(t) < R_{th}), \mathbb{I}(I_k^u(t) < I_{th}))}}. \quad (14)$$

The proposed reward function incorporates an exponential penalty term $2^{\max(\mathbb{I}(R_k^u(t) < R_{th}), \mathbb{I}(I_k^u(t) < I_{th}))}$, halving the reward upon Quality of Service (QoS) threshold violations. This mechanism effectively integrates QoS constraints into the reinforcement learning framework, guiding the system to prioritize actions that mitigate violations.

D. Computational Complexity Analysis

This section discusses the computational complexity of the proposed algorithms. 1) *Clustering Algorithm*: It consists of DWAHC and dynamic load balancing. DWAHC computes the density-weighted distance for each user and merges clusters iteratively with a complexity of $O(K^2)$, while the dynamic load balancing algorithm adjusts the load of U clusters with a complexity of $O(2 \cdot U \cdot K)$. When U is much smaller than K , the overall complexity is approximated as $O(K^2)$. 2) *Optimization Algorithm Training Phase*: The backpropagation process in the fully connected neural network has a complexity of $O(|A| \cdot |S| \cdot n)$, where $|S| = 3 \cdot U + K$, $|A| = 7 \cdot U + 3 \cdot K$, and n is the total number of hidden layer nodes. Assuming E episodes and batch size C , the overall training complexity is $O(E \cdot C \cdot t \cdot U \cdot |A| \cdot |S| \cdot n)$. 3) *Optimization Algorithm Testing Phase*: Since UAVs are only responsible for inference decisions, the complexity is $O((10U + 4K + 1) \cdot n \cdot t \cdot U)$.

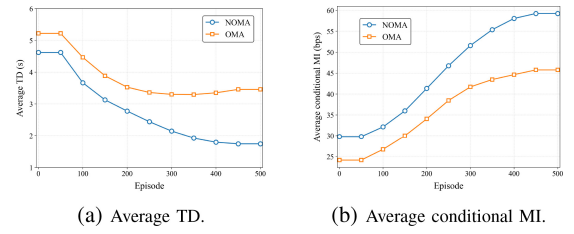


Fig. 3. System performance across multiple access modes.

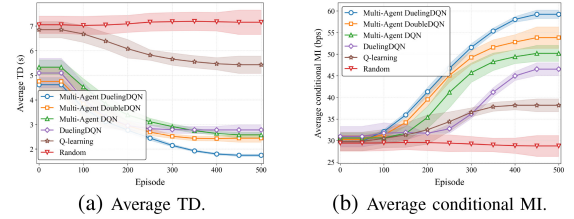


Fig. 4. Performance with optimization algorithms.

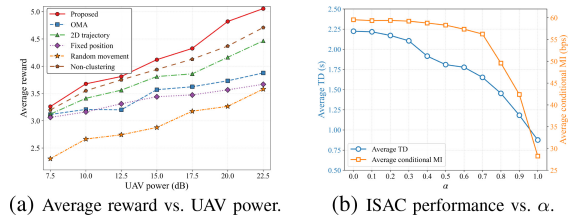


Fig. 5. Performance Analysis of ISAC.

IV. PERFORMANCE EVALUATIONS

The simulations cover a 500 m^2 area with UAVs operating at altitudes of 20 m–150 m. Ground users exhibit random mobility with a maximum speed of 0.5 m/s, while UAVs maintain a default velocity of 5 m/s. Key parameters include [12], [16]: $B = 4 \text{ MHz}$, $\sigma^2 = -100 \text{ dBm}$, $T_\nu = 5 \text{ us}$, $\Delta f = 0.25 \text{ MHz}$, $P_{\max} = 29 \text{ dBm}$, $R_{th} = 0.15 \text{ kb/s}$, $I_{th} = 10 \text{ bps}$, and $\alpha = 0.5$.

Fig. 3 compares the ISAC performance of different multiple access schemes. The proposed NOMA scheme outperforms the OMA scheme in both average TD and conditional MI. Initially, both schemes perform similarly due to comparable resource setups. However, as iterations progress, the advantage of the NOMA scheme becomes apparent by allowing multiple users to share the same resources, thereby enhancing spectral efficiency and improving ISAC performance. In contrast, the OMA scheme's orthogonal resource allocation leads to fragmentation and inefficiency, especially in high-density scenarios with limited resources.

Fig. 4 compares the proposed multi-agent DuelingDQN algorithm with baseline methods. The proposed method achieves reduced average TD and higher average conditional MI, attributed to its dueling architecture mitigating Q-value overestimation and collaborative multi-agent framework enabling UAVs to share global information.

Fig. 5(a) compares the proposed algorithm with multiple baselines under identical NOMA-ISAC configurations, where each baseline differs from the proposed algorithm in specific technical components. The proposed algorithm consistently

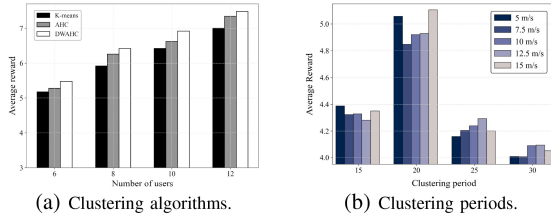


Fig. 6. Average reward vs. Clustering parameters.

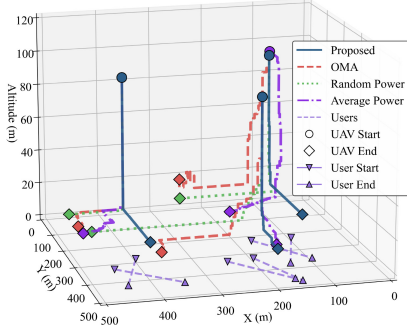


Fig. 7. 3D trajectory optimization.

outperforms all alternatives across varying UAV transmit power levels. Compared to the OMA scheme, which employs orthogonal resource allocation, the NOMA-based design enhances spectral efficiency through power-domain multiplexing. The proposed algorithm adaptively adjusts UAV altitude via 3D optimization to improve spatial resource utilization, surpassing the 2D trajectory algorithm. The fixed-position algorithm suffers from inadequate dynamic mobility for user tracking, while the random movement algorithm induces suboptimal trajectory coordination. The non-clustering algorithm degrades ISAC performance due to the absence of effective dynamic user clustering. Fig. 5(b) shows the ISAC performance under different values of α . As α increases, both the average TD and average conditional MI gradually decrease, as the system increasingly prioritizes the optimization of average TD. This indicates that the system can flexibly balance TD and conditional MI by adjusting the weight of α , thereby optimizing different objectives.

Fig. 6(a) compares the DWAHC algorithm with K-means and AHC across varying user scales, showing improvements of 7.2% and 3.2% in average reward, respectively. This performance boost stems from the density-weighted distance metric, which dynamically adjusts to user distributions, enabling more balanced clusters even in uneven scenarios. Fig. 6(b) illustrates the effect of clustering periodicity on system performance. The proposed algorithm achieves peak average reward at $M = 20$ slots, with a 14.7%–22.8% gain over other clustering periods, confirming the need for periodic reclustering. The algorithm also maintains stability with standard deviations below 0.1 across different UAV velocities, demonstrating consistent performance regardless of speed.

Fig. 7 illustrates the trajectory optimization performance of different algorithms, with identical user trajectories used across all algorithms for comparison. As shown, UAVs under the proposed algorithm first descend to an appropriate altitude, then travel in straight lines toward their assigned user

clusters, and finally remain stationary at optimized positions. In contrast, although the comparative algorithms show some tracking capability, they do not fully cover all user clusters. Experimental results confirm that the proposed trajectory optimization framework effectively tracks user mobility while ensuring continuous service coverage.

V. CONCLUSION

This letter investigates dynamic user clustering, trajectory planning, and power allocation optimization in NOMA-ISAC empowered UAV networks. A two phase density-aware clustering algorithm is developed for efficient user association, complemented by a multi-agent DuelingDQN framework for joint trajectory and power control. Experimental results demonstrate superior performance over existing baseline methods in reducing TD and improving conditional MI. The proposed scheme enables robust UAV-ground interactions through NOMA-ISAC transmission, achieving balanced ISAC performance in dynamic environments.

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